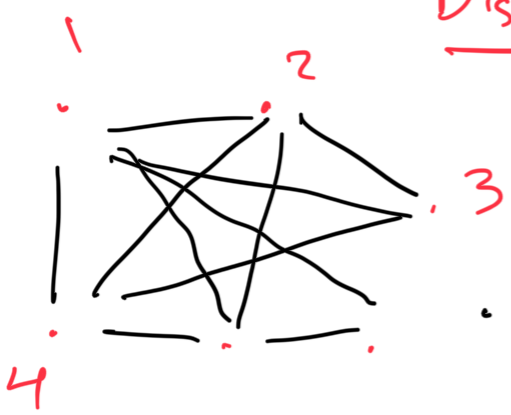
Distributed Computing

9/8/2026

Tuesday



- N nodes, complete graph
- $\binom{n}{2}$ edges in total
- Bidirectional

Leader Election

- Finish fast (minimize steps)
- Less communication : Minimize amount of msgs sent
- Synchronous clocks . Total msgs sent = $n(n-1)$
- Anonymous / Labelled nodes

↳ No deterministic algorithm

- Receive → Compute → Send

  == 1 round / step

$P \subseteq$ BPP — Bounded Probabilistic Polynomial

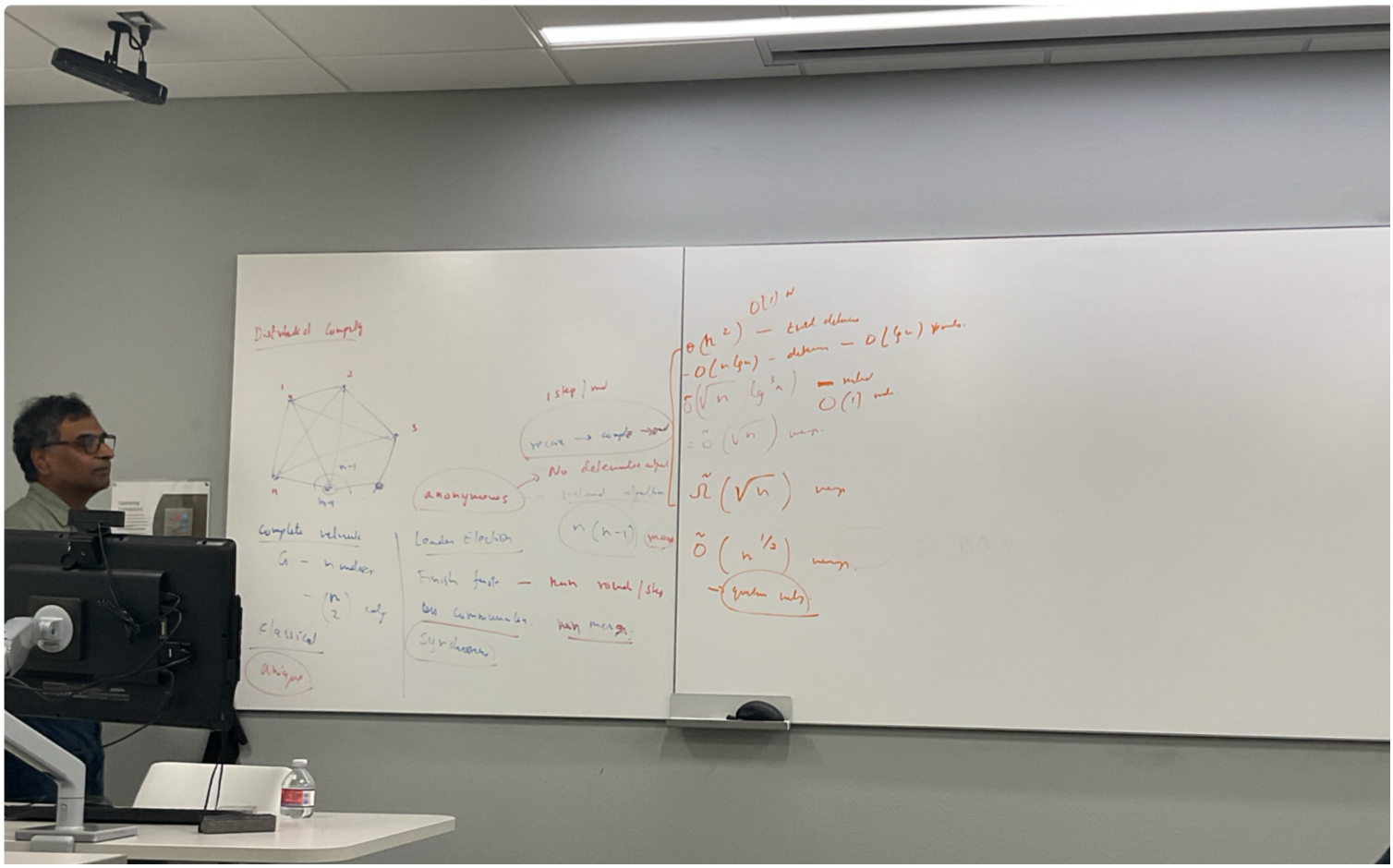
Polynomial Deterministic

$\subseteq$ BQP — Bounded Quantum Polynomial

Assuming non-anonymous, unique

ids, $\tilde{O}(n^2)$ - trivial classical solution, $O(1)$ rounds

$O(n \log n)$ - deterministic w/ $O(\log n)$ rounds





(1) Subsample candidates

$\rightarrow \frac{2 \log n}{n}$

(2) Each candidate sends ID to $\sqrt{n \log n}$ random nodes

$E(\underbrace{\#\text{ candidates}}_{X}) = \frac{2 \log n}{n} \times n$

$= 2 \log n$

$$\Pr[X < 1] = \Pr[X < (1-\delta)\, 2\log n)$$

[from Chernoff bounds]

$$= e^{-2\log n \left(1 - \frac{1}{2\log n}\right)^2}$$

$$= \frac{1}{n^2}$$

$$\Pr[X \geq 1] = 1 - \frac{1}{n^2}$$

(3) Referee round:

Among the received messages,

winner is the minimum id node

(4) Leader is the winner for all rounds.

Atleast $\underline{1}$ leader $\in$ Show

Almost $\underline{1}$ leader $\in$ Show

$$P(X \leq \underline{1} \text{ leader})$$

$$\leq \left(1 - \frac{\sqrt{n \log n}}{n}\right)^{\sqrt{n \log n}}$$

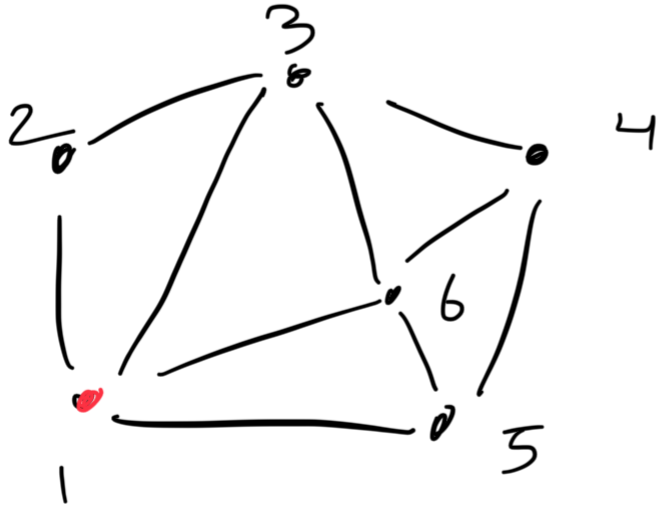
$$\leq e^{-\frac{\sqrt{n \log n}}{n}\sqrt{n \log n}}$$

$$\leq e^{-\frac{x \log n}{x}}$$

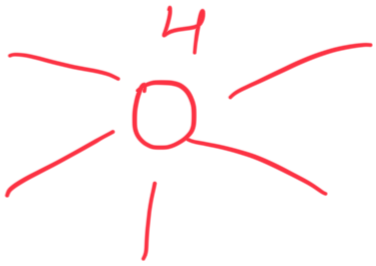
$$= \frac{1}{n}$$

Markov Chains

Random walk in an undirected graph



(1,0,0,0,0,0)



$\frac{1}{deg(n)}$

$deg(n)$

Random walk : moving to some neighboring node with uniform probability $\frac{1}{\deg(u)}$.

Consider,



$P_0 = (1, 0, 0)$

$P_1 = (0, \frac{1}{2}, \frac{1}{2})$

$P_2 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$

$$P_t = (p_t^1, p_t^2, p_t^3)$$

- Show "law of large numbers" with Chebyshev inequality.